

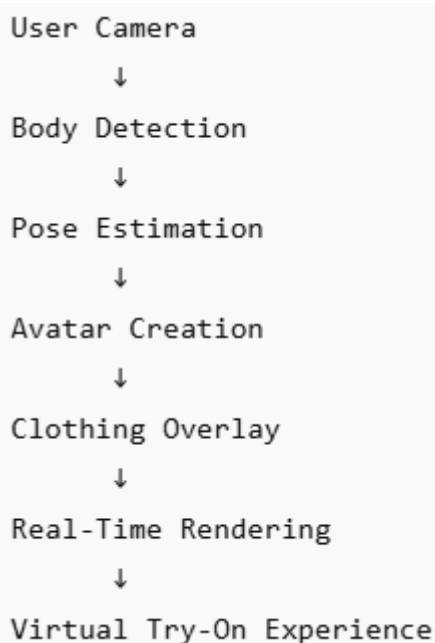
AR/VR Virtual Try-On System Research Report for Raritone

Introduction

The fashion industry is rapidly adopting Augmented Reality (AR) and Virtual Reality (VR) technologies to enhance customer experiences and reduce the uncertainty associated with online shopping. One of the most promising applications of these technologies is the Virtual Try-On System, which allows users to visualize clothing, footwear, accessories, and other fashion products on themselves before making a purchase. By combining computer vision, artificial intelligence, and real-time rendering technologies, virtual try-on systems create an interactive shopping experience that bridges the gap between physical and digital retail environments.

For Raritone, implementing an AR/VR-based virtual try-on solution has the potential to improve customer engagement, increase purchase confidence, and reduce product return rates. This research focuses on understanding the technologies, frameworks, and methodologies involved in developing an effective virtual try-on experience.

Understanding the Virtual Try-On Workflow



Camera Input and User Detection

A virtual try-on system operates through a sequence of interconnected processes that work together to provide a realistic and responsive user experience. The process begins when the user accesses the application and enables the device camera. The camera captures real-time images or video streams, which are then analyzed using computer vision algorithms.

Body Tracking and Pose Estimation

The system identifies key body landmarks and tracks user movements using pose estimation techniques. These landmarks represent important

body joints and positions, allowing the application to understand the user's posture and orientation. Based on this information, a digital avatar or body representation is generated, enabling the placement of virtual garments in a realistic manner.

Garment Alignment and Rendering

Once the clothing asset is loaded into the system, alignment algorithms ensure that the garment matches the user's body position and movement. The clothing must dynamically adjust as the user changes posture, turns, or moves. Finally, the rendering engine combines the real-world camera feed with the virtual garment, creating a seamless augmented reality experience in real time.

Importance of Accuracy

The effectiveness of a virtual try-on system depends heavily on accurate body tracking, realistic garment placement, and low-latency rendering. Any inaccuracies in these components can negatively impact the user experience.

AR/VR Technologies Explored

Several technologies were explored to understand their suitability for developing Raritone's virtual try-on platform.

WebAR

WebAR represents one of the most accessible approaches to augmented reality. Unlike traditional AR applications that require installation, WebAR experiences run directly within a web browser. This allows users to access virtual try-on features instantly through a URL, making the onboarding process significantly easier. WebAR is particularly attractive for e-commerce applications because it eliminates the need for app downloads. However, browser-based limitations can impact performance and reduce the complexity of graphical experiences.

ARKit

ARKit is Apple's augmented reality framework designed specifically for iOS devices. It provides advanced features such as face tracking, body tracking, motion capture, and environmental understanding. ARKit is known for its accuracy and smooth performance due to Apple's close integration between hardware and software. For fashion applications, ARKit can be used to create highly responsive virtual try-on experiences with precise body and facial tracking capabilities.

ARCore

ARCore is Google's equivalent framework for Android devices. It enables motion tracking, environmental awareness, and depth sensing capabilities. ARCore allows

developers to create immersive AR experiences across a wide range of Android smartphones. Its ability to understand real-world environments makes it valuable for applications that require stable object placement and realistic interactions.

Unity AR Foundation

Unity AR Foundation serves as a bridge between ARKit and ARCore, allowing developers to build applications using a single codebase that can be deployed on both Android and iOS devices. This significantly reduces development complexity and ensures consistent functionality across platforms. Unity's extensive ecosystem, large community, and compatibility with third-party tools make it one of the most practical choices for AR application development.

Three.js

Three.js is a JavaScript-based 3D graphics library that enables developers to create interactive 3D experiences directly within web browsers. It is particularly useful for WebAR implementations where browser compatibility and lightweight deployment are important considerations. Three.js can be used to render virtual garments, avatars, and interactive product visualizations.

Blender

Blender is an open-source 3D modeling and animation software widely used for creating digital assets. In the context of virtual try-on systems, Blender can be used to design clothing models, create avatars, edit textures, and prepare assets for integration into AR environments. Its flexibility and extensive feature set make it an essential tool for fashion-focused AR development.

AI Models and Computer Vision Technologies

Artificial Intelligence plays a critical role in enabling realistic virtual try-on experiences. Accurate body tracking and garment alignment rely heavily on machine learning models capable of understanding human posture and movement.

MediaPipe Pose

MediaPipe Pose is a lightweight and efficient framework developed for real-time body landmark detection. It identifies key points on the human body and tracks them continuously as the user moves. Due to its speed and low computational requirements, MediaPipe Pose is well



suiting for mobile-based AR applications. It provides a practical solution for implementing clothing alignment and basic virtual fitting functionality.

OpenPose

OpenPose is a more advanced pose estimation framework capable of detecting multiple body joints and tracking multiple individuals simultaneously. Its accuracy and flexibility make it suitable for complex virtual try-on systems where detailed body understanding is required. However, it typically requires greater computational resources compared to MediaPipe.

SMPL (Skinned Multi-Person Linear Model)

SMPL represents a significant advancement in digital human modeling. Unlike traditional pose estimation frameworks, SMPL generates realistic three-dimensional body representations based on body measurements and pose information. This enables more accurate clothing fitting, body size estimation, and personalized avatar creation. SMPL is particularly valuable for future implementations where size recommendations and realistic garment simulation become priorities.

Role of AI in Virtual Try-On

The combination of pose estimation models and body modeling technologies enables a virtual try-on system to move beyond simple visual overlays and deliver highly personalized experiences.

Competitor Analysis

Several leading companies have successfully implemented AR-based virtual try-on solutions, providing valuable insights for Raritone.

Snapchat

Snapchat has become one of the most recognized platforms for augmented reality experiences. Its AR try-on features demonstrate highly responsive face tracking and object placement capabilities. Users can interact with accessories, eyewear, and fashion products in real time, highlighting the importance of low-latency tracking and user-friendly interfaces.

Gucci

Gucci has utilized augmented reality technology to allow customers to virtually try on footwear before purchasing. The experience emphasizes realism and visual quality, helping users better understand how products appear in real-world scenarios. Gucci's implementation demonstrates the value of integrating AR into luxury fashion retail.

Nike

Nike's virtual try-on solution focuses primarily on footwear visualization. By leveraging advanced tracking technologies, Nike provides users with a realistic representation of shoe fit and appearance. The application illustrates how AR can reduce uncertainty during the purchasing process and improve customer confidence.

Zeekit

Zeekit pioneered virtual clothing try-on technology by enabling users to visualize garments on digital representations of themselves. Its approach emphasizes personalized fitting and size recommendations, highlighting the growing importance of AI-driven body measurement systems in modern fashion technology.

Key Insights

The success of these platforms indicates that users increasingly expect interactive and personalized shopping experiences. Their implementations provide valuable references for the development of Raritone's virtual try-on ecosystem.

Industry Trends

AI-Based Body Measurement Systems

The virtual fashion industry is evolving rapidly as advancements in artificial intelligence and augmented reality continue to improve user experiences. One major trend is the integration of AI-powered body measurement systems that automatically estimate user dimensions and recommend appropriate clothing sizes. This capability helps reduce return rates and improves customer satisfaction.

Personalized Digital Avatars

Another emerging trend is the use of personalized avatars that accurately represent individual users. Rather than relying solely on camera overlays, future virtual try-on systems are expected to utilize realistic digital twins that can be customized and reused across multiple shopping experiences.

Real-Time Cloth Simulation

Real-time cloth simulation is also gaining attention as computational power increases. Advanced physics engines can simulate fabric behavior, allowing garments to respond naturally to body movement. This significantly enhances realism and provides users with a more accurate representation of how clothing would behave in reality.

Browser-Based AR Experiences and Generative AI

Browser-based AR experiences are becoming increasingly popular because they remove barriers to entry and allow users to access virtual try-on functionality instantly. Additionally, generative AI technologies are beginning to influence fashion

visualization by enabling rapid garment customization and personalized design previews.

Future Outlook

These trends indicate that future virtual try-on platforms will become more immersive, intelligent, and personalized.

Recommendations for Raritone

Development Platform

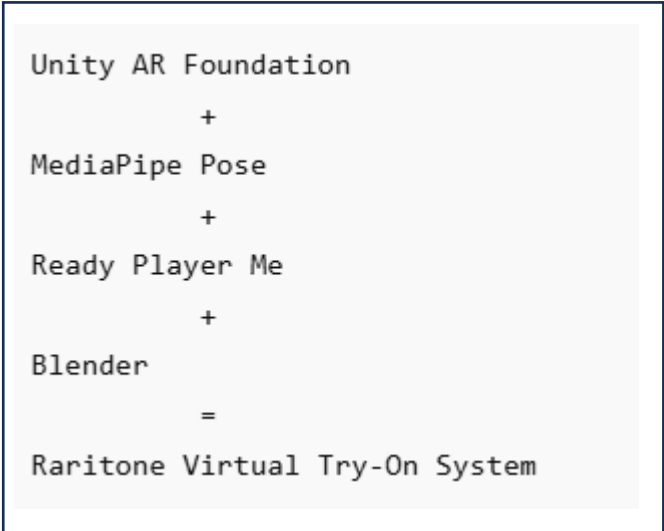
Based on the research conducted, Unity AR Foundation is recommended as the primary development platform due to its ability to support both Android and iOS devices through a single codebase. This approach provides flexibility, scalability, and efficient resource utilization during development.

Body Tracking Solution

For body tracking and pose estimation, MediaPipe Pose offers an excellent balance between accuracy and performance. Its lightweight architecture makes it suitable for mobile deployment while providing sufficient tracking capabilities for initial virtual try-on implementations.

Avatar and Asset Creation

Ready Player Me or similar avatar-generation platforms can accelerate development by providing customizable avatars that integrate easily with AR environments. Blender should be utilized for creating and modifying clothing assets, ensuring that garments can be optimized for real-time rendering.



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Unity AR Foundation
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MediaPipe Pose
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Future Enhancements

As the platform evolves, SMPL-based body modeling and AI-powered measurement systems should be explored to improve fitting accuracy and personalization. These technologies will enable Raritone to deliver more realistic and customer-centric virtual shopping experiences.

Conclusion

The research demonstrates that AR and AI technologies have transformed the way consumers interact with fashion products in digital environments. Virtual try-on systems combine computer vision, body tracking, 3D modeling, and real-time rendering to create engaging shopping experiences that increase user confidence and satisfaction.

Among the technologies explored, Unity AR Foundation, MediaPipe Pose, and Blender emerge as the most practical combination for developing an initial prototype for Raritone. Future enhancements involving personalized avatars, body measurement estimation, and cloth simulation can further strengthen the platform's capabilities. By adopting these technologies strategically, Raritone can position itself at the forefront of next-generation fashion retail experiences.